VANISHING FOR THE HOMOLOGY SHEAF OF A COMPLEX OF QUASI-COHERENT MODULES ON THE ETALE AND FPPF SITES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-vanishing-adequate (source lines 1615-1639); the proof environment beginning after the author's following remark paragraph (source lines 1641-1647) is its own proof and closes at source line 1703.

Editorial note. The remark between the statement and the proof (introducing adequate modules and pointing to the Adequate Modules chapter) is part of the surrounding narrative, not of the counted statement or proof, and is not retained.

Editorial note. The proof invokes one result of the same chapter by label (retained as context with the generator's transitive expansion) and two cross-chapter results (Cohomology on Sites, Descent), each declared as a tagged source pointer.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: the vanishing of the etale and fppf cohomology of the homology sheaf of a complex of quasi-coherent modules at affine objects.

Allowed standard prerequisites (editorial).

- Restriction of the big etale site to a small one. For f : U -> X in the big etale site one has i_f o pi_X = f as morphisms of ringed sites U_etale -> X_etale, and restriction along i_f is exact.. It identifies H_etale|_{U_etale} with the degree-one homology of f^*F_bullet at source lines 1650-1658.
- The quasi-coherent review lemma. The presheaf Y -> Gamma(Y, f^*F) satisfies the fpqc (hence fppf and etale) sheaf condition (the author's lemma-review-quasi-coherent, retained as context).. It gives that H_etale is an fppf sheaf and hence H_etale = epsilon_{X,*}H_fppf at source lines 1672-1682.
- Cech vanishing criterion for a collection of coverings. If Cech cohomology vanishes for a suitable collection of coverings, then the sheaf cohomology vanishes (Cohomology on Sites, declared as a tagged source pointer).. It reduces the vanishing to the displayed Cech comparison at source lines 1685-1691.
- Cech vanishing for quasi-coherent modules on a standard covering. The standard covering Cech complex of a quasi-coherent module has no higher

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cohomology (Descent, declared as a tagged source pointer).. It finishes the proof at source lines 1700-1702.

Source reference. Original lemma 03F9: sources/stacks-chow/tags.txt, source lines 4484–4484. Cross-chapter pointer whose target carries the canonical tag 03F9 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03FJ: sources/stacks-chow/tags.txt, source lines 4494–4494. Cross-chapter pointer whose target carries the canonical tag 03FJ in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cross-chapter pointer whose target carries the canonical tag 04W8 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04TS: sources/stacks-chow/tags.txt, source lines 6265–6265. Cross-chapter pointer whose target carries the canonical tag 04TS in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DEW: sources/stacks-chow/tags.txt, source lines 16816–16816. Cross-chapter pointer whose target carries the canonical tag 0DEW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03DK: sources/stacks-chow/tags.txt, source lines 4423–4423. Cross-chapter pointer whose target carries the canonical tag 03DK in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter pointer whose target carries the canonical tag 03CG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter pointer whose target carries the canonical tag 03A0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module.*

(1) *The rule*
$$\mathcal{F}^a : (Spaces/X)_{étale} \longrightarrow Ab, \quad (f : Y \to X) \longmapsto \Gamma(Y, f^*\mathcal{F})$$
satisfies the sheaf condition for fpqc and a fortiori fppf and étale coverings,
(2) $\mathcal{F}^a = \pi_X^*\mathcal{F}$ *on* $(Spaces/X)_{étale}$,
(3) $\mathcal{F}^a = a_X^*\mathcal{F}$ *on* $(Spaces/X)_{fppf}$,
(4) *the rule $\mathcal{F} \mapsto \mathcal{F}^a$ defines an equivalence between quasi-coherent $\mathcal{O}_X$-modules and quasi-coherent modules on $((Spaces/X)_{étale}, \mathcal{O})$,*

(5) *the rule* $\mathcal{F} \mapsto \mathcal{F}^a$ *defines an equivalence between quasi-coherent* $\mathcal{O}_X$*-modules and quasi-coherent modules on* $((Spaces/X)_{fppf}, \mathcal{O})$,
(6) *we have* $\epsilon_{X,*}a_X^*\mathcal{F} = \pi_X^*\mathcal{F}$ *and* $a_{X,*}a_X^*\mathcal{F} = \mathcal{F}$,
(7) *we have* $R^i\epsilon_{X,*}(a_X^*\mathcal{F}) = 0$ *and* $R^ia_{X,*}(a_X^*\mathcal{F}) = 0$ *for* $i > 0$.

Proof. Part (1) is a consequence of fppf descent of quasi-coherent modules. Namely, suppose that $\{f_i : U_i \to U\}$ is an fpqc covering in $(Spaces/X)_{étale}$. Denote $g : U \to X$ the structure morphism. Suppose that we have a family of sections $s_i \in \Gamma(U_i, f_i^*g^*\mathcal{F})$ such that $s_i|_{U_i \times_U U_j} = s_j|_{U_i \times_U U_j}$. We have to find the correspond section $s \in \Gamma(U, g^*\mathcal{F})$. We can reinterpret the s_i as a family of maps $\varphi_i : f_i^*\mathcal{O}_U = \mathcal{O}_{U_i} \to f_i^*g^*\mathcal{F}$ compatible with the canonical descent data associated to the quasi-coherent sheaves $\mathcal{O}_U$ and $g^*\mathcal{F}$ on U. Hence by Descent on Spaces, Proposition 04W8 we see that we may (uniquely) descend these to a map $\mathcal{O}_U \to g^*\mathcal{F}$ which gives us our section s.

We will deduce (2) – (7) from the corresponding statement for schemes. Choose an étale covering $\{X_i \to X\}_{i \in I}$ where each X_i is a scheme. Observe that $X_i \times_X X_j$ is a scheme too. This covering induces a covering of the final object in each of the three sites $(Spaces/X)_{fppf}$, $(Spaces/X)_{étale}$, and $X_{étale}$. Hence we see that the category of sheaves on these sites are equivalent to descent data for these coverings, see Sites, Lemma 04TS. Parts (2), (3) are local (because we have the glueing statement). Being quasi-coherent is a local property, hence parts (4), (5) are local. Clearly (6) and (7) are local. It follows that it suffices to prove parts (2) – (7) of the lemma when X is a scheme.

Assume X is a scheme. The embeddings $(Sch/X)_{étale} \subset (Spaces/X)_{étale}$ and $(Sch/X)_{fppf} \subset (Spaces/X)_{fppf}$ determine equivalences of ringed topoi by Lemma 1.1. We conclude that (2) – (7) follows from the case of schemes. Étale Cohomology, Lemma 0DEW. To transport the property of being quasi-coherent via this equivalence use that being quasi-coherent is an intrinsic property of modules as explained in Modules on Sites, Section 03DK. Some minor details omitted. □

1. Comparing big and small topoi

Lemma 1.1. *Let* S *be a scheme. Let* $\tau \in \{étale, fppf, ph\}$ *(add more here). The inclusion functor*

$$(Sch/S)_\tau \longrightarrow (Spaces/S)_\tau$$

is a special cocontinuous functor (Sites, Definition 03CG) and hence identifies topoi.

Proof. The conditions of Sites, Lemma 03A0 are immediately verified as our functor is fully faithful and as every algebraic space has an étale covering by schemes. □

Lemma 1.2. *Let* S *be a scheme. Let* X *be an algebraic space over* S*. Let* $\mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3$ *be a complex of quasi-coherent* $\mathcal{O}_X$*-modules. Set*

$$\mathcal{H}_{étale} = \mathrm{Ker}(\pi_X^*\mathcal{F}_2 \to \pi_X^*\mathcal{F}_3)/\mathrm{Im}(\pi_X^*\mathcal{F}_1 \to \pi_X^*\mathcal{F}_2)$$

on $(Spaces/X)_{étale}$ *and set*

$$\mathcal{H}_{fppf} = \mathrm{Ker}(a_X^*\mathcal{F}_2 \to a_X^*\mathcal{F}_3)/\mathrm{Im}(a_X^*\mathcal{F}_1 \to a_X^*\mathcal{F}_2)$$

on $(Spaces/X)_{fppf}$*. Then* $\mathcal{H}_{étale} = \epsilon_{X,*}\mathcal{H}_{fppf}$ *and*

$$H^p_{étale}(U, \mathcal{H}_{étale}) = H^p_{fppf}(U, \mathcal{H}_{fppf}) = 0$$

for $p > 0$ *and any affine object* U *of* $(Spaces/X)_{étale}$.

Proof. For any object $f : U \to X$ of $(Spaces/X)_{étale}$ consider the restriction $\mathcal{H}_{étale}|_{U_{étale}}$ of $\mathcal{H}_{étale}$ to $U_{étale}$ via the functor $i_f^* = i_f^{-1}$ discussed in Section 1. The sheaf $\mathcal{H}_{étale}|_{U_{étale}}$ is equal to the homology of complex $f^*\mathcal{F}_\bullet$ in degree 1. This is true because $i_f \circ \pi_X = f$ as morphisms of ringed sites $U_{étale} \to X_{étale}$. In particular we see that $\mathcal{H}_{étale}|_{U_{étale}}$ is a quasi-coherent $\mathcal{O}_U$-module. Next, let $g : V \to U$ be a flat morphism in $(Spaces/X)_{étale}$. Since

$$i_{f\circ g}^* \circ \pi_X^* = (f \circ g)^* = g^* \circ f^*$$

as morphisms of sites $V_{étale} \to X_{étale}$ and since g is flat hence g^* is exact, we obtain

$$\mathcal{H}_{étale}|_{V_{étale}} = g^* \left(\mathcal{H}_{étale}|_{U_{étale}}\right)$$

With these preparations we are ready to prove the lemma.

Let $\mathcal{U} = \{g_i : U_i \to U\}_{i\in I}$ be an fppf covering with $f : U \to X$ as above. The sheaf property holds for $\mathcal{H}_{étale}$ and the covering $\mathcal{U}$ by (1) of Lemma 0.1 applied to $\mathcal{H}_{étale}|_{U_{étale}}$ and the above. Therefore we see that $\mathcal{H}_{étale}$ is already an fppf sheaf and this means that $\mathcal{H}_{fppf}$ is equal to $\mathcal{H}_{étale}$ as a presheaf. In particular $\mathcal{H}_{étale} = \epsilon_{X,*}\mathcal{H}_{fppf}$.

Finally, to prove the vanishing, we use Cohomology on Sites, Lemma 03F9. We let $\mathcal{B}$ be the affine objects of $(Spaces/X)_{fppf}$ and we let Cov be the set of finite fppf coverings $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ with U, U_i affine. We have

$$\check{H}^p(\mathcal{U}, \mathcal{H}_{étale}) = \check{H}^p(\mathcal{U}, (\mathcal{H}_{étale}|_{U_{étale}})^a)$$

because the values of $\mathcal{H}_{étale}$ on the affine schemes $U_{i_0} \times_U \ldots \times_U U_{i_p}$ flat over U agree with the values of the pullback of the quasi-coherent module $\mathcal{H}_{étale}|_{U_{étale}}$ by the first paragraph. Hence we obtain vanishing by Descent, Lemma 03FJ. This finishes the proof. □